

A novel and secured email classification and emotion detection using hybrid deep neural network

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ABSTRACT

Compared to other social media data, email data differs from it in various topic-specific ways, including extensive replies, formal language, significant length disparities, high levels of anomalies, and indirect linkages. In this paper, the creation of a potent and computationally effective classifier to categorize spam and ham email documents is proposed. To assess and validate spam texts, this paper employs a variety of data mining-based classification approaches. On the benchmark Enron dataset, which is open to the public, tests were run. The final 7 Enron datasets were created by combining the six different types of Enron datasets that we had acquired. We preprocess the dataset at an early stage to exclude any useless phrases. This method falls under several categories, including Logistic Regression (LR), Convolutional Neural Networks (CNN), Random Forests (RF), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and suggested Deep Neural Networks (DNN). Using Bidirectional Long Short-Term Memory (BiLSTM), email documents may be screened for spam and labeled as such. In performance comparisons, DNN-BiLSTM outperforms other classifiers in terms of accuracy on all seven Enron datasets. In comparison to other machine learning classifiers, the findings demonstrate that DNN-BiLSTM and Convolutional Neural Networks can categorize spam with 96.39 % and 98.69 % accuracy, respectively. The report also covers the dangers of managing cloud data and the security problems that might occur. To safeguard data in the cloud while maintaining privacy, hybrid encryption is examined in this white paper. In the AES-Rabit hybrid encryption system, the symmetric session key exchange-based Rabit technique is combined with the benefits of the AES algorithm for faster data encryption.

1. Introduction

An email has grown in importance and effectiveness as a tool for both personal and professional communication. Some email users misuse the service by flooding recipients inboxes with spam and computer worms. There were an estimated 54 billion spam emails sent daily in 2014, according to statistics (Venmathi et al., 2023). Email servers are overloaded by spam, which uses up storage space and bandwidth on the network. To address these issues, email filtering is therefore a crucial step. Spam will be located and quarantined by filtering.

These emails, which are often referred to as spam, are sent by several businesses and untrusted individuals and are used to advertise things, sell them, or deliver URLs. Therefore, they do not have to pay to send emails, as individuals or businesses send bulk emails to other receivers. Spam aids in the sale of goods and information thievery. They either

employ this information to lure people into illegal activity or utilize it to carry out financial transactions. Today's computer age makes spam lethal since it allows us to gather people's sensitive and crucial information. Because modern computers are so quick, spam has become a significant issue. The major spam problem must be handled with extreme caution because individuals still lack the necessary understanding to grasp it even today. Using techniques like machine learning and email classification, one can deal with problems like spam (Kaspersky 2021). Numerous studies presently use decision trees, naïve bays, and support vector machines as classifiers to categorize spam content. These methods are based on data mining and machine learning. The researchers employed a range of evolutionary methods, such as genetic algorithms, particle population optimization, and principal component analysis, to hone the functional space of the dataset.

Numerous email scams have been reported to them, as well as

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experienced. The rise in events necessitates the use of text email forensics. Email analytics fall into two categories. Use the email body as well as the header information. Our research focuses on Type II emails, which have text-based bodies. An individual author with a distinctive writing style creates each email (Grover et al., 2021). By analyzing the writing style, we can determine who wrote the email. Early in 1887, Thomas Corwin Mendenhall started the study by examining the writing styles of Beacon, Marlowe, and Shakespeare. Each person writes differently. To accurately predict authorship, a model that takes into account writing style is necessary (NacerEddine et al., 2019).

On the other hand, phishing and legitimate categories frequently demand a large amount of data for DNN model training. To do an in-depth study, it is challenging to gather emails with personal data on servers (Fard and Mahoor, 2022). Additional privacy concerns must be made because of the data that the model collects and processes. It has been harder, or even impossible, to use centralized learning to manage sensitive textual data in emails, particularly with the rise of legislative constraints like GDPR. In addition, people are reluctant to share data due to privacy concerns, and trained models may not perform as expected when put to use if sufficient training data is not available. When a system detects a fresh sample it has never seen before, it may mistakenly identify a legal email as phishing, leading to a false positive (Reka et al., 2023).

The body of the email will be used in this study to introduce a novel way of categorizing spam and ham emails. To increase computing performance, this method makes advantage of several method complexity techniques. Beginning with the preprocessing and cleaning of email formats including the elimination of stopwords and unnecessary data techniques then move on to semantic modelling for feature compression before using feature reduction and feature selection strategies. Only storing the most crucial features is the major objective (Green et al., 2021). The proposed architecture and the sequential process both incorporate the provided methods and approaches.

Because e-mail is so inexpensive, its popularity on the Internet has elevated it to the top spot among all forms of communication. However, this capability does not give spammers the ability to send large amounts of spam to make money, steal online account information and personal information, or infect recipients' computers with viruses or malware. The benefit of this is. This is due to the fact that existing email systems do not have a strong identity and security system and instead employ Simple Mail Transfer Protocol (SMTP), which does not validate the origin of email. Anyone can spoof and send spam over the currently available SMTP, which can be used inappropriately. Users regularly get a certain quantity of spam from unknown senders in their mailboxes (Mheidly et al., 2020).

The primary contributions of this research are:

- A challenge that hasn't been fully solved yet is measuring semantic relationships and similarities of WordNet ontologies to semantically reduce email functionality.
- As far as aware no research has been done to demonstrate the effectiveness of this feature selection strategy for categorizing spam.
- To fill in these knowledge gaps, provide 12 sentiment classification models and Markov chain models for examining time-oriented sentiment trends.
- In case of studies, use the analysis recommended in Enron's emails to identify emotional distinctions between guilty and innocent actors.
- The categorization method for email author attributes in this paper uses deep neural networks (DNN) and LSTM.

1.1. Challenges

The problem at hand involves finding ways to make the main telecommunications firms that are supporting this paper's automated email classification and sorting processes function better. The suggested model

that would replace the existing model ought to have a higher or equivalent F1 score and a better classification rate. The ability to incorporate the model into the present production environment should also be offered.

The infrastructure and all email required for classifier evaluation will be provided by Telco. Experts in the relevant fields will assess the suggested framework in a test setting and contrast it with the most recent rule-based classifiers. If the suggested model outperforms the company's existing classifiers and provides additional business value, it will be implemented into the manufacturing process. Swedish email taxonomy development was made possible by the study question. Although though it belongs to a telecoms firm, the framework might be used as the starting point for additional studies on multi-label emails and semantic analysis.

2. Literature survey

Email filtering frequently makes use of map classification techniques (Giovannelli et al., 2021). Documents are given pre-defined class labels (such as "ham" or "spam") based on probabilities given by the labeled text training set. Naive Bayes (NB) (Mheidly et al., 2020), Artificial Neural Networks (ANN) (Annamalai & Muthiah, 2022), K nearest neighbours (Grahlow et al., 2020), logistic regression (Grundmann et al., 2021), C4.5 classifiers (Jiménez et al., 2021), RBF, etc. Networks (Grundmann et al., 2021), Multilayer Perceptron (Annamalai & Muthiah, 2022), AdaBoost (Grahlow et al., 2020, Grundmann et al., 2021), SVM (Support Vector Machine) (Jiménez et al., 2021), and Random Forest (RF) (Jiménez et al., 2021) are some of the techniques used for spam filtering.

In Canal et al. (2022), utilizing RBF, provide an ANN-based method (Radial Basis Function Network). Put in place the SENTINEL anti-spam filter. The filter takes into account the author's writing style while extracting natural language characteristics from the email text. With the use of the CSDMC2010 dataset, compare the performance of two meta-algorithms: RF, SVM, NB, and ADABOOST (Nilabar et al., 2023).

For a long time, researchers have researched centralized phishing email detection. ML- and DNN-based email security can be effectively solved by the aforementioned methods, however, centralized data processing may jeopardize the confidentiality and privacy of routine email communications (Manikandan et al., 2023). As a result, any efforts to detect phishing emails must use privacy architecture. Unfortunately, no studies have been done on detecting phishing emails using distributed learning. To compare various settings, they used distributions of training data and email user counts that varied. As a result, make sure that the performance of your model declines noticeably as the number of clients rises (Bouktif et al., 2020).

An approach to phishing detection that is based on content was suggested by the authors of Karnati et al. (2023). To categorize phishing emails, the authors use random forests. Phishing and junk email were divided into categories. Using relevant feature extraction and increased prediction accuracy, to enhance the phishing email classifier. By selecting the optimum kernel settings, the accuracy can be further increased. The analysis of emails was done manually by Yang et al. (2019). The writers uncovered and decoded fake messages that were sent via SMTP, looked at IP addresses, tracked down places, and made a timeline of all the activities. To corroborate the activities stated in the timeline, also investigated the server logs, but it was a time-consuming process that required a significant amount of email data.

There has been substantial progress in the detection of phishing emails as they have emerged as a danger. Quan et al. (2021) using 18 distinct hybrid signatures, including email header structure, email URL information, and email script signatures, (Elngar et al., 2021) were able to recognize phishing emails. When using SVM, classification accuracy can reach 95 %. Support Vector Machines, Random Forests, Logistic Regression, Naive Bayes, and Voting Perceptron were used by Rastenis et al. (2021) (2017) to categorize emails as phishing or ham. They

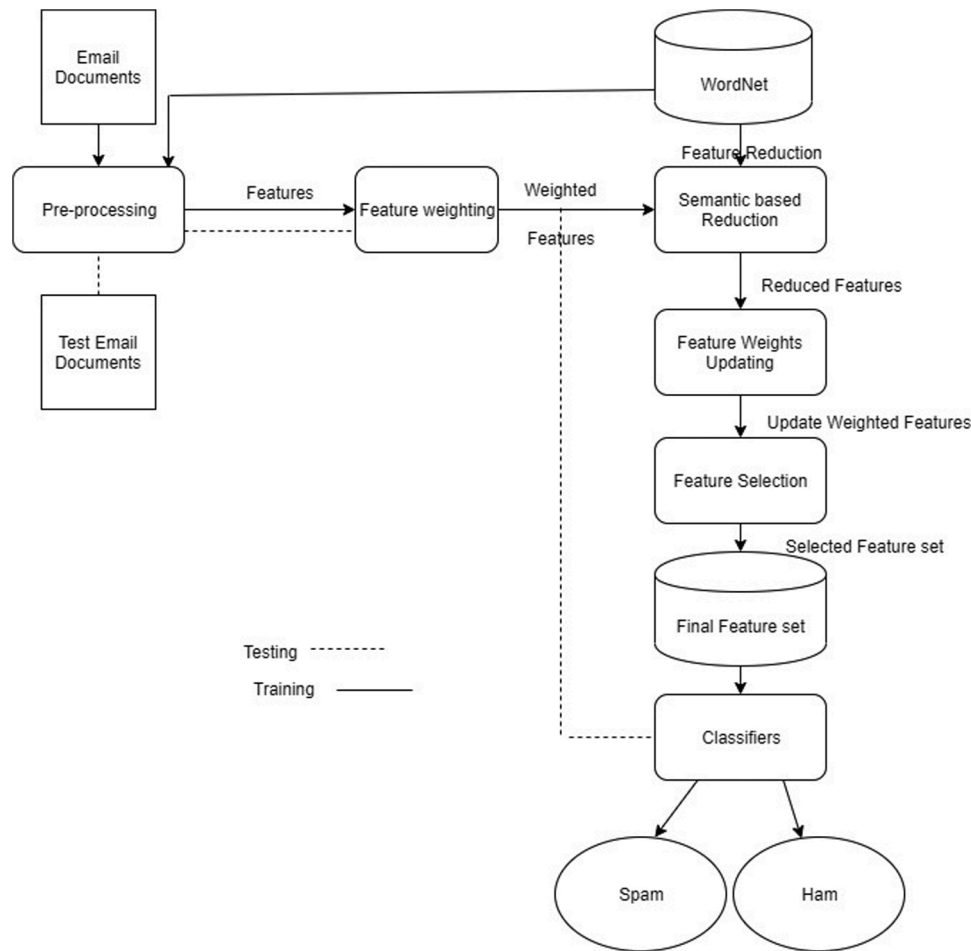


Fig. 1. The proposed system architecture.

achieved up to 99.87 % accuracy when compared to these classifiers (Gopalan et al., 2023). In Verma et al. (2020), developed an intelligent classification model to identify phishing emails by calculating each email's "phishing email weight" using the concept of "phishing word weight." Also employed were WordNet ontology and text morphological analysis. With J48 and Random Forest, were able to obtain 98.4 % accuracy (Pathak et al., 2022). To categorize emails into spam and phishing emails based on their content, (Rameshkumar et al., 2019) provides a method using text classification. Themis is a brand-new model for phishing email detection proposed by Babalola et al. (2019) that uses multi-level vectors in a recurrent CNN model. Email headers, bodies, character levels, and word levels are all simultaneously modeled using these. A very high accuracy of 99.84 % was attained in the classification process (Kumar et al., 2021). The classification of phishing emails was explored by Kumar et al. (2021) using NLP.

Using data from the Enron scandal, describe how non-spam papers are classified in this section. Big data security was a theme that the author (Khanduzi and Sangaiah, 2023) noticed when conducting the study. We studied how to deal with spam with students using the Enron dataset, which included spam and ham files, to conclude big data security challenges (Mahmood et al., 2023). The authors of this study (Mahmood et al., 2023) choose to model and provide appropriate receivers using Enron's staff assignment information. The Enron email dataset and the STDG simulator tool are used by the authors to build a synthetic dataset in this research (Manikandan & PonniBala, 2023, Kalaivani and Seetharaman, 2022) to detect threading and knowledge discovery.

2.1. Research gap

This investigation has two research holes. Initially, research is done on several subjects, including models that understand natural language and classifiers that take advantage of word relationships in time series that are necessary for properly classifying emails.

The effects of stock, classification label aggregation, LSTM network size, and depth selection on classification performance have not been well studied. This study's primary topic of inquiry served as its inspiration.

Second, numerous machine-learning techniques for document classification have been extensively studied. However, building discrete document groups cannot be done with email since it is informal, allows for fair competition in terms of social hierarchy, promotes personal closure, and has the potential to be emotional. This is especially true given that there is a paucity of studies on email.

3. Proposed approach

The suggested approach divides emails into two groups: ham and spam contain several components that scale back the characteristics. Fig. 1 illustrates the proposed architecture. Training and testing are their two components. Four key elements make up the training phase. Reduce and classify features based on feature weight. The testing procedure involves feature weighting, classification, and preprocessing. Shrinking the modules is the most crucial aspect of the architecture. The semantic discovery, weight function, and feature selection procedures make up this module's three proposed processes. In comparison to past work, the minify module is an advancement. Sections 3.1 and 3.2 each explain the

preprocessing and feature weight modules, while Section 4 explains the feature reduction module.

3.1. Pre-processing

In trials involving text mining, preprocessing is a crucial stage. Creating well-formed data from unusual input is the largest issue in text mining. To turn unusual data into well-formed data, there are various phases involved. In addition to word count, morphological analysis, tokenization, trimming, and stop words, we also use other processes. A document's total word count is determined by word count. Switching relevant data elements is a process known as tokenization. Before using stop words to eliminate frequent terms like prepositions and conventions (a, is, etc.), pruning eliminates unnecessary words to improve structural data. Here is a method.

Tokens are extracted by the preprocessing program, and unnecessary tokens like numbers and symbols are also eliminated. The email's body and subject line are used to extract the token. Eliminate stop words next. WordNet is used as an English dictionary and only meaningful terms are taken into account to arrange more data. WordNet may be viewed as a sizable library of English vocabulary that organizes English words into synsets.

Analyzing morphology is not used in this module. For the function's meaning to be maintained, do this. Instead of morphological analysis, morphology is used in this case. Every characteristic that has the same root is regarded as a marker in morphology. The email document "d," following preprocessing, can be expressed as follows:

$$d_i = (t_1, w_1), (t_2, w_2) \dots (t_n, w_n) \quad (1)$$

The text is normalized and made ready for analysis as part of the data preprocessing stage using natural language processing techniques. As listed below, there are various steps involved.

Tokenization: Tokenization in natural language processing is the division of the raw text into parts. Tokenizing documents as words adhere to a set of predetermined rules. Python's SpaCy module is used to carry out the tokenization stage.

Stop Words Removal: Frequently used words like "a" and "the" add noise to text data and are unimportant in the context of the email. They can filter the text to analyze to remove certain terms, which are referred to as stop words. To remove non-words from text, use the Python package "NLTK".

Punctuation Removal: Punctuation marks, such as a period (.), comma (,), and square brackets, divide sentences and make their meanings clear. Use the NLTK library to get rid of punctuation.

To convert the remaining characters to lowercase, it first takes off non-alphabetic characters such as digits, punctuation, etc. Tokenization is a process that separates sentences into smaller word chunks by extracting tokens from the input text. It eliminates non-words like "the," "a," "is," etc. It changes words in different tenses and plurals to their base or dictionary form, depending on the base type. As a result of the aforementioned stages, words with fewer than two letters are frequently eliminated since they are only unfinished words or fragments of words that have no clear meaning. To create continuous strings using lexical analysis, finally reassemble these clean, normalized tokens.

Additionally, we need to encode these extracted texts as numeric feature vectors to input the email data to the DL model for training. To do this, we train a mapping from input strings to vector representations using embedding layers. Layer embedding can be learned in one environment and used in another. Client collaboration on skewed local training data is another need for distributed model training. Therefore, each client employs a global word embedding strategy that makes use of a pre-trained embedding model rather than creating a single embedding layer for word representations.

3.2. Feature weighting

Each extracted feature's value is recorded by this module. Thus, using the Tom Frequency/Inverse Document Frequency approach, a weight "w" is given to each phrase "t" (TF-IDF).

The number of times the term "t" appears in email document "d" is the unit used in the term frequency Eq. (2).

$$tf(t, d) = \frac{f_d(t)}{\max(f_d(t))} \quad (2)$$

where $f_d(t)$ is the frequency of phrase "t" in email "d."

A term's relevance is estimated using IDF (Inverse Document Frequency). A term's rarity in a document is determined using Eq. (3).

$$IDF(t) = \log\left(\frac{N}{df_t}\right) \quad (3)$$

where "N" is the total number of emails and "df t" is the number of emails that include the word "t." To obtain TF-IDF, multiply Eqs. (2) and (3).

$$W = tf(t, d) * IDF(t) \quad (4)$$

3.2.1. Feature extraction

To cluster and categorize the emails from each author, the suggested model uses machine learning techniques. The development of a feature set is a crucial component of achieving this. The actual writing style of each author is based on the particular ways in which they create and display their knowledge. These details serve as the foundation for the extraction of style measurement features for each author taken into account in the email dataset. The email body's text serves as the source for these functionalities. The statistical measure TF-IDF vectorizer, shown in Fig. 4, is used to determine weights by calculating how similar a word is to each document that is being taken into consideration. TF-IDF is a multipurpose method that transforms text into useful numerical or vector representations. A significant aspect of the text is represented by the depiction. TF-IDF is an occurrence-based vector format that normalizes word occurrence to size and the number of times it appears in other authors using a combination of two metrics. One way to distinguish emails from other writers is to utilize this specific email body as an identifying feature. An n-gram analysis of the words and a count vectorizer were used to handle the cleaned email dataset. By employing a specific similarity, the reciprocal and reciprocal frequency is determined. Elements are mathematically represented as cosines and measured in Euclid.

$$tf(w, e) = \frac{freq(count(w), e)}{|e|} = \log(1 + freq(w, e)) \quad (5)$$

$$idf(w, e) = (\log|E|) \frac{1}{cont(e \in E : W \in e)} \quad (6)$$

$$tf(w, e, E) = tf(w, e) * idf(w, E) \quad (7)$$

Freq(w, e) returns the term frequency in email e, whereas tf(w, e) returns the normalized frequency of term t in email e. The length of the email body is |This| words. In circumstances when a term is often used and appears in several documents, the formula idf(w,e) yields an inverse document frequency of 0; otherwise, it yields an inverse document frequency of 1. The email for the entire dataset is represented by |E|. The count (e:E: W) function displays the number of emails that have the letter "w" in the email body. A more significant email phrase is eqn. The digits 5 and 6 return the number 7.

This quality is taken into account in our work.

- Word count in the text.
- Word count in the unique text.
- The words' typical length.

- The text's character count.
- How many text elements are not words.
- The amount of punctuation used in the text.
- Word capitalization rate.
- Length of case words in the title

3.2.2. Feature weights update

Once the feature set has been decreased, terms are given new weights based on the semantic similarity metric that has been used. The distance between two words is produced once the semantic measure has been calculated. To assess how similar two words are to one another, this distance is measured against a specified threshold; if it is less than the threshold, the word's weight is modified to the new weight.

$$F = (w_i, w_j) \times (1 - dist_{ij}) \quad (8)$$

where w_i and w_j are the weights given to terms i and j , and $dist_{ij}$ is a measure of two words. The method of updating feature weights is schematically depicted in Fig. 5.

3.3. Feature selection

The process of choosing the features that are most valuable out of all the features is called feature selection. Using this strategy, all features that have a big impact are chosen, and all elements that have a minimal impact are eliminated. The correlation between word characteristics and categorizations is used in chi-square. A category's likelihood of occurring is referred to as its probability, whereas a characteristic refers to how frequently that feature occurs. When dependent, seek correlation values and avoid applying FST. Otherwise, look for correlations between features and categories. The procedure calculates the characteristics' rankings using a ranking mechanism. A property's value can be removed if it falls below a certain level. Ranker is used after the entropy-based FST called "Information Gain," or IG. The volume of data made available for text category feature items is defined by IG. To determine how important lexical terms are to categorization, IG is determined based on how much a phrase is used to categorize information. The FST method is an expansion of the Relief feature selection algorithm from 1994. Data classification with several classes is supported by Relief. In the training set, Relief chooses samples at random. Search for k close hits in the same class and k close misses in each separate level after that. In this manner, modify the feature weights, and carry out this procedure n times to raise all of the feature weights.

3.4. Experiment 1: DNN-BiLSTM classification

3.4.1. Deep neural network (DNN)

The information processing paradigm known as an artificial neural network (ANN) is based on how biological nerve systems function when the brain analyses information. It is made up of densely interconnected processing units, known as neurons, which cooperate to address a particular issue. A network of artificial neurons made up of several layers is referred to as a "deep neural network" or "deep learning." The utilization of rich layers, which have been suggested for stylistic characters, is the main component of our strategy. Since each author has a distinctive writing style, improve this method. Every concealed layer has the following mathematical form:

$$y = Wx + b \quad (9)$$

In a deflection matrix, where X is the input data (for the first layer) and W and b are the weights, the learned parameters taken into account during training are represented. It is referred to as a thick layer since every neuron is coupled to every other neuron. Each node's activation function has an impact on the network's output, making it a crucial component of all neural networks. As they were successful for us, we decided to use ReLu and softmax for the project.

$$z = f(b + e \cdot w) = f\left(b + \sum_{i=1}^n e_i \cdot w_i\right) \quad (10)$$

Eq. (10) illustrates how the network nodes work. Through the activation function and bias b , the input's weighted sum, e , is transmitted.

An activation function is used to activate the data once it has been input into each connected node of the neural network. Creators are tasked with predicting data in the final hierarchy.

To train a model, weights, and biases are applied iteratively to the dataset. Gather probabilities for every author then calculate gradients for each parameter. A gradient, which is a vector, is used to indicate the loss function's local minimum. Categorical cross-entropy loss is the employed loss function. For network optimization, the global minimum is applied. We employ the Adam optimizer, which was created especially to operate with deep neural networks, to calculate the learning rate for each parameter. Because it does a decent job of handling sparse gradients, employ this optimizer for noisy issues. The dataset contemplated for research has the traits that are most typical and discovered that employing the Adam optimizer yields positive outcomes.

Through the use of triple cross-validation, our approach builds three consecutive neural networks and tests their correctness. Accuracy is further increased by increasing the folds and epochs. The artificial neural network (ANN) approaches use as computational models and incorporate intelligence into the suggested method since it is influenced by the characteristics of biological neural networks. A particular sort of artificial neural network known as a feed-forward neural network is described by directed graphs, which transfer the various system of information along edges without creating loops from one node to another (FFN). Three or more layers of FFN-type multi-layer perceptron (MLP) models, each with a significant number of neurons or mathematical symbolic units, make up the input layer, one or more hidden layers, and the output layer. The selection of the hyperparameters determines how many hidden layers are used. Each layer's neurons are completely interconnected, and data is sent from one layer to the next. The MLP mathematical definition is provided here. $O: R^m \times R^n$, where m is the input vector and $x = x_1, x_2, \dots, x_m$, n are the input and output vector sizes, respectively, of the value x . Each hidden layer's calculation is explained by the mathematical definitions that follow.

$$h_i(x) = f(w_i^T x + b_i) \quad (11)$$

One of them is $i: R^{d_i}$, where d_i is the input, and f is a nonlinear activation function such as a sigmoid (value range $[0, 1]$) or tangent (value range $[-1, 1]$). To address issues with multi-class classification, MLP models employ the softmax function as a non-linear activation function. The softmax function outputs the probability for each class and chooses the biggest probability value to offer a more precise estimate. The following formulae describe the sigmoid, tangent, and softmax activation functions:

$$sigmoid = \frac{1}{1 + e^{-x}} \quad (12)$$

$$tangent = \frac{e^{2x} - 1}{e^{2x} + 1} \quad (13)$$

$$softmax(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}} \quad (14)$$

where x defines an input.

The three-layer MLP with a soft tmax function at the output layer is the same as the multiclass logistic regression model. MLP is often represented as for several hidden layers.

$$H(x) = H_l(H_{l-1}(H_{l-2}(\dots(H_1(x))))) \quad (15)$$

Stacking hidden layers using this method is sometimes referred to as a DNN (Deep Neural Network). Fig. 7 illustrates a deep neural network

(DNN) design with a single hidden layer. An overview of a deep neural network is shown in Fig. 6. Output $O = O(1), O(2), \dots, O(3)$, O from input $x = x_1, x_2, \dots, x_{m1}, x_m$ (4).

An employ DNN as a more sophisticated version of the conventional FFN, with ReLU nonlinear activation functions for each hidden layer. This lessens the issue of misleading slopes and missing states. ReLU has a benefit over other nonlinear activation functions in that it allows for quicker and simpler training of MLP models with more hidden layers. The biggest neuron determines the neural network's breadth, and the hidden layer determines its depth. The cleverness of our strategy is to efficiently maximize deep learning by modeling the loss function and ReLU.

1) Loss function: Achieving good performance while modeling MLPs requires choosing the best parameters. The loss function is a preliminary step in this. Use a loss function to determine the difference between the expected and desired values defined mathematically as:

$$d(t, p) = \|t - p\|_2^2 \quad (16)$$

where p stands for the predicted value and t for the desired value.

Use the negative log probability as follows for multiclass classification with target class t and probability distribution $p(pd)$:

$$d(t, p(pd)) = -\log p(pd)_t \quad (17)$$

Use the negative log probability as follows for multiclass classification with target class t and probability distribution $p(pd)$:

$$\text{loss}(i_n, o_n) = \frac{1}{n} \sum_{i=1}^n d(o_i, f(i_i)) \quad (18)$$

A neural network needs to minimise its loss function in order to perform better. The definition of a loss function is:

$$\text{Train}_{i_o}(\theta) = L_{i_o}(\theta) = \frac{1}{n} \sum_{i=1}^n d(o_i, f_{\theta}(i_i)) \quad (19)$$

where the loss function's minimization is denoted by

$$\min_{\theta} L(\theta) \quad (20)$$

There are many optimization methods; choose gradient descent, which is the most popular. Gradient descent computes and modifies parameters iteratively while adhering to the following rules:

$$\theta^{\text{new}} = \theta_{\text{old}} - \alpha \nabla_{\theta} L(\theta) \quad (21)$$

where the selection of the hyperparameters determines the choice of, which stands for the learning rate. The back-propagation or error back-propagation algorithm is used to determine the derivative of L . Chain rule computation is used in backpropagation to determine R_d , which reduces the loss function $L_i o()$. To find the minimum, however, most neural networks employ stochastic gradient descent (SGD), a back-propagation extension. Mini-batches of training samples are used by SGD. Instead of utilising im^i, om^i for the whole training set, the training samples io in this case are picked at random. The following are the SGD updating rules:

$$\theta^{\text{new}} = \theta_{\text{old}} - \alpha \nabla_{\theta} J(\theta; im^i, om^i) \quad (22)$$

2) Rectified Linear Unit (ReLU): This unit has good proficiency and has been shown to speed learning. ReLU has significantly reduced the vanishing and exploding gradient issue in neural network history. It is the fastest and most economical method for training massive quantities of data when compared to conventional non-linear activation functions like sigmoid and tangent. The ReLU mathematical formula is:

$$f(x) = \max(0, x) \quad (23)$$

where x defines input.

3.4.2. Long-short memory (LSTM)

The LSTM is a novel and scalable learning model for sequence data. In many ways, LSTMs perform better than CNNs and RNNs. RNNs are good at handling short-term dependencies, but they can't recognize long-term dependencies or input events that are widely spread in time or place, and context. Vanishing gradients is a difficulty with RNNs. When the matrix's weights are too small, learning slows down or ceases altogether due to the vanishing gradient problem. Contrarily, the learning may diverge, which is the so-called "gradient explosion problem," if the weight of this matrix is high and the slope is excessively steep. This has been the main driving force for the usage of LSTM models, which include new structures termed memory cells that are shielded from outside influences by peripheral gates.

Three dependencies are necessary for the memory unit to function. Based on the previous time step, the previous concealed state, the output of the prior unit, and the input of the current step, information is stored in memory (new input information). A storage space serves the purpose of keeping something valuable for a while. This is accomplished by applying a deactivation function to the memory cell unit, which has no cumulative gradient reduction factor. The logistics function is used to gate data coming into and leaving the unit. How many function values are utilized to compute the output is decided by the output gate. The input gate controls how much fresh data enters the cell. How effectively the previous value is kept in the cell is determined by the forget gate. The most popular models for categorical data training are recurrent neural networks (RNNs). Training with traditional RNNs continues to fail. Before using LSTM to solve the issue, briefly explain the RNN form and loss problem in this part.

LSTM networks are used as storage in place of the original RNN model's hidden hierarchical nodes. (The cell state e key is used to hold historical data.) The sigmoid function and the pointwise product operation can be utilized with three gate layouts to update or remove cell state data. Fig. 6 depicts the inner LSTM cells as forget gates, input gates, and output gates, in that order. The processing of sequence information by LSTM networks is cumulative and linear to prevent gradient loss and develop long-term knowledge (Babalola et al., 2019). It can be applied to learning.

The forget gate's equation is

$$f_t = \sigma(w_f * [h_{t-1}, x_t] + b_f) \quad (24)$$

Where f_t is the forget gate's output value, h_{t-1} is the output value at the previous instant, x_t is the input value at the present time, and w_f and b_f are the sigmoid function's weight matrix and deflection vector. Avoid considering the door. $[h_{t-1}, x_t]$ is the connection matrix for h_{t-1} and x_t .

The input gate's equations are

$$i_t = \sigma(w_i * [h_{t-1}, x_t] + b_i) \quad (25)$$

$$k_t = \tanh(w_k * [h_{t-1}, x_t] + b_k) \quad (26)$$

Where i_t and k_t are the input gate's output values and w_i and b_i are the weight matrix and bias vector of the sigmoid function, respectively. The letters w_k and b_k , respectively, stand for the weight matrix and deflection vector of the input gate's tanh function.

Here is the output gate formula:

$$O_t = \sigma(w_o * [h_{t-1}, x_t] + b_o) \quad (27)$$

$$h_t = O_t * \tanh(C_t) \quad (28)$$

Where w_o and b_o are the weight matrix and deflection vector in the sigmoid function of the output gate, respectively, and h_t is the current instantaneous output value. O_t is the output value of the output gate.

The updated cell is in the following condition:

$$C_t = f_t * C_{t-1} + i_t * k_t \quad (29)$$

Here, C_t stands for the condition of the cell in the present, and C_{t-1} for the state of the cell in the past.

3.4.3. Bi-directional long short-term memory (Bi-LSTM) model

Long short-term memory was first postulated by Hochreiter and Schmidhuber (Canal et al., 2022) in 1997. The last layer is a dense layer that accepts input from the final output LSTM layer and has softmax activation. Change the parameter settings, the number of layers, the activation functions, the learning rates, and the loss functions to see what works best for detecting network attacks using one to six layers (L1-L6). Select layer 5 (L5) for further experimental evaluation because, according to our experimental findings, LSTM Layer 5 can categorize the majority of spam mail items with respectable detection accuracy.

The default LSTM is limited to accessing the context before each particular time step. However, continuous sensory data from machine condition monitoring systems exhibit significant time dependence. It makes sense to take future circumstances into account. As a result, a bidirectional LSTM is used here. Using two distinct hidden layers and forward feeding to the same output layer, bidirectional LSTMs are capable of handling bidirectional sequential data, including forward and backward approaches.

3.4.4. DNN with bidirectional LSTMs

Deep architectures have recently been demonstrated to be effective at learning representations. So it makes sense to build a deep LSTM neural network by stacking several LSTM layers. A core idea of deep neural networks is the necessity that model inputs pass through a lot of nonlinear layers. A deep LSTM model's sensor input may flow through several LSTM layers. As seen in Fig. 4c, the concealed output of one LSTM layer is not only transmitted over time but also employed as input data for the following LSTM layer. A layer's output sequence is passed on to the following layer. In bidirectional LSTM architecture, an input sequence made up of the output sequences of the prior layer's forward and backward layers is given to each hidden layer.

The forget gate layer (Fig. 4a) employs the sigmoid function to select what information the LSTM strips from the cell state while displaying the input data and data from previously hidden layers (leave 1, 0 for removal). It is calculated as

$$f_t = (W_f \cdot [ht1, xt] + b_f) \quad (30)$$

Information stored by the LSTM in the cell state is determined by the input/update gate layer (Fig. 9). Input gate layers employ the sigmoid function to determine which data has to be updated, whereas Tanh layers recommend new vectors to add to the cell state. Ten LSTMs update the cell's state by forgetting information it has determined to forget and updating it with new vector values. It was determined that it was equal to

$$(W_i \cdot [ht1, xt] + b_i) \quad (31)$$

and that Ct was equal to

$$C_t = (\tanh(W_c \cdot [ht1, xt] + b_c)) \quad (32)$$

Output layer (Fig. 4c): This layer applies the sigmoid function to the cell LSTM output to ascertain what the output is. The outcome passes through the Tanh layer (values ranging from -1 to 1). It just outputs the data that it chooses to send to the following neuron. figured out as follows:

$$O_t = [W_o[ht1, xt] + b_o] \text{ and } ht = [ot * \tanh(C_t)] \quad (33)$$

Algorithm:

Step 1: Input dataset (Enron Email)

Step 2: Initialize the kernel function

Step 3: Input layer

Layer: lstm_3 (LSTM)

Data: Train data

Input Size: (2395,1)

Output size: (None, 2395,100)

Step 4: Hidden layers:

4.1: Convolutional Layer:

Layer: Convolution 1D

Input Data: lstm_3 (LSTM)

Input Size: (None,2395,100)

Output size: (None, 2395,32)

4.2: Maxpooling Layer:

Layer: Maxpooling 1D

Input Data: conv1d_2(Conv1D)

Input Size: (None,2395,32)

Output size: (None, 1197,32)

4.3 LSTM Layer:

Layer: LSTM

Input Data: max_pooling_2 (MaxPooling1)

Input Size: (None,1197,32)

Output size: (None, 50)

4.4 Dropout Layer:

Input Layer: Dropout layer

Input Data: lstm_4(LSTM)

Input Size: (None,50)

Output size: (None, 50)

4.5 Dense Layer:

Layer: Dense layer

Input Data: dropout_4(Dropout)

Input Size: (None,50)

Output size: (None, 1)

Step 5: Output Layer:

Input data:

Input Size: (None, 1)

Output Size:

Step 6: Fitting Data:

Input Data: Train data

Output: Accuracy

Step 7: End

3.5. Experiment 2: emotion detection

This section aims to provide an unsupervised strategy to identify the 12 emotions stated in Section 2 in the absence of labeled training data and specifically suggested by the Enron emails to the test. The public Enron email corpus contains information on 158 people and roughly 2 million emails. The psychological condition of people who are impacted by interactions inside the company is shown by the emotional state of users as inferred from these emails. The emotions of criminals and innocent bystanders were contrasted in our study (the ultimate criterion for litigation). Ascertain that the two groups are distinct. 22 former Enron workers were found guilty, while 21 were found not guilty. They perform comparable roles within the firm as managers or officers. It comprises the total number of emails sent and received as well as the days on which they were sent and received. Sentiment analysis for each email is performed at the sentence level.

There are several drawbacks to this strategy. First, if sentiment terms

Table 1
Enron datasets.

Datasets	Ham-instance	Spam-Instance	Total instances
Enron-1	3673	1502	5174
Enron-2	4363	1498	5859
Enron-3	4015	1504	5514
Enron-4	1503	4502	6001
Enron-5	1504	3678	5177
Enron-6	1505	4504	6002
Combine Enron	16563	17188	33727

Table 2
Number of instances of Enron datasets.

Datasets	Total instances
Enron-1	5174
Enron-2	5859
Enron-3	5514
Enron-4	6001
Enron-5	5177
Enron-6	6002
Combine Enron	33727

are not tagged, dictionary-based approaches are incorrect. Any structure that flips the meaning will fall short. For instance, the negative emotion of worry is expressed by the phrase "can't rest." When there are no emotional words, feelings might nonetheless manifest in particular circumstances. Second, despite the implied tie between fact words and emotion terms and the dislocated emotional connection caused by semantic similarity, a dictionary-based grouping of pure synonyms lacks these connections. For instance, the word "excited" with a good connotation is primarily negative but has the synonym "aame." Additionally, "after-widget" may be derogatory and cruel, but none of the synonyms in the thesaurus are sentimental. The root of the word "wisdom" has been shown to have a favorable effect in vocabulary-based investigations. Third, it's difficult to create rule-based solutions, and they don't cover all the bases. Exclamation points and a few rhetorical questions, for instance, might be used to express strong feelings. To completely understand grammatical rules and language examples, huge datasets and supervised machine learning is a necessity.

In light of all these drawbacks, suggest a sentiment identification technique that combines fundamental grammar with words that have been labelled as having a certain emotion and the WordNet vocabulary. The following phases make up the proposed method's processing of the raw text.

Step 1: Label each sentence. Target part-of-speech tags are determined because adjectives, adverbs, nouns, and verbs all transmit

Table 3
Pre-processing of Enron datasets.

Pre-Processing Steps	Number of Attributes						
	Enron-1	Enron-2	Enron-3	Enron-4	Enron-5	Enron-6	Combined Enron
Word count	50559	148916	53874	69530	42287	71599	155125
Stemmer	35625	141395	33866	48971	27337	47160	109469
Tokenizer	30854	22216	29728	44178	23696	41840	94508
Pruning	8765	10852	13032	12221	9335	11813	33527
Stop word	8531	10614	12790	11982	9102	11577	33274

Table 4
Accuracy of the classifiers using the datasets from Enron.

Classifier	Enron-1	Enron-2	Enron-3	Enron-4	Enron-5	Enron-6	Combined Enron
LR	90.94 %	94.84 %	62.95 %	88.79 %	91.5 %	97.13 %	96.16 %
CNN	93.73 %	96 %	94.93 %	96.39 %	94.39 %	94.44 %	96.39 %
Random Forest	87.54 %	86.39 %	86.84 %	86.49 %	88.6 %	87.98 %	88.69 %
RNN	89.24 %	89.58 %	88.29 %	92.13 %	89.61 %	86.46 %	87.93 %
LSTM	85.67 %	80.09 %	90.56 %	89.32 %	87.56 %	79.9 %	78.66 %
DNN-BiLSTM (Proposed)	98.54 %	98.39 %	99.84 %	99.49 %	98.6 %	98.98 %	98.69 %

Table 5
DNN-BiLSTM classifier's confusion matrix while using the Enron datasets.

Actual vs Predicted	Enron-1		Enron-2		Enron-3		Enron-4		Enron-5		Enron-6		Combine Enron	
	Ham	Spam	Ham	Spam	Ham	Spam	Ham	Spam	Ham	Spam	Ham	Spam	Ham	Spam
Ham	3604	72	4349	16	4010	6	1291	213	1431	73	1381	123	16153	250
Spam	60	1444	200	1300	173	1331	2	4501	4	3675	3	4501	204	16199

Table 6
Measures of the best DNN-BiLSTM classifier performance using several Enron datasets.

Number of Measures	Datasets						
	Enron-1	Enron-2	Enron-3	Enron-4	Enron-5	Enron-6	Combined Enron
TP rate	98.11	99.7	99.92	85.95	95.29	91.95	98.56
FP rate	3.89	13.26	11.42	2	0.07	0.04	1.2
Precision	98.44	95.66	95.93	102	99.88	99.95	98.83
Specificity	96.15	86.78	88.62	102	99.97	100	98.84
F-Measure	98.27	97.64	97.88	92.45	97.53	95.78	98.69

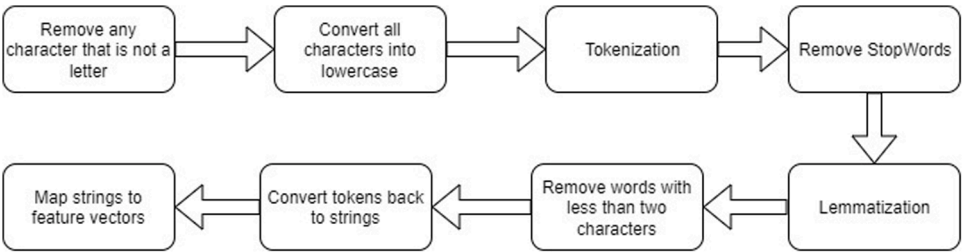


Fig. 2. Text mining pipeline for email data.

Text Sequence	Lexical Analysis Result
CONFIRM YOUR DELIVERY DETAILS	confirm delivery detail
1st Reminder - Invitation to 2021 UI GreenMet...	reminder invitation world univers
Purchase order537893	purchase order
Password Notification Notification For Your ...	password notification notification passcode

Fig. 3. Sample result of the preprocessing.

emotional information. The part-of-speech markers for some words are several. I like to read, for instance, use the verb "like" to express interest and a favourable attitude. Students were endangered on the child's behalf, despite it being a preposition that does not convey any emotion. Get candidate groupings of the target words.

Step 2: Conduct a sentiment word search and use a thesaurus to find the candidate group's marked sentiment terms. Language index and word count (LIWC) dictionary word lists are used to grade dictionaries on a positive and negative scale. The word is eliminated from the candidate set once the sentiment term has been verified.

Step 3: In the remaining possible groups, use the thesaurus to locate words associated with emotional vocabulary. The search for empathetic words is over now.

Step 4: Add up all the occurrences of each sentiment to determine the overall sentiment of an email. This is done after identifying the

sentiment language and its lexicon for all sentences. In light of this, obtain the sentiment distribution for each email.

Emotions can be influenced or changed by negative language construction. To effectively recognize these changes, however, machine learning and training are needed. Since the impacts of potential alterations are thought to be eliminated after aggregation. The relationship between unfavorable feelings and emotions, however, was examined. Neg-x, where x is multiplied by the unlawfulness of the statement, indicates that a sentence is unlawful. We discovered the following after doing our analysis: (a) Many people employ the syntaxes "or not" and "whether or not." (b) Sentences are long because they are not properly divided by terminal characters (e.g., in certain emails, sentences are connected with "{"); lengthen.

3.6. Dual encryption algorithm

In today's modern technological era, when individuals utilize remote servers and the internet to deliver and store data and applications from the cloud, cloud technology is incredibly significant and advantageous constructed using common encryption methods. Advanced Encryption Standard (AES) is a relatively novel technique.

Nowadays AES is likely to be used as the most effective and powerful algorithm But presumed that in some cases AES is not enough for security, So for more security purposes we have experimented with AES

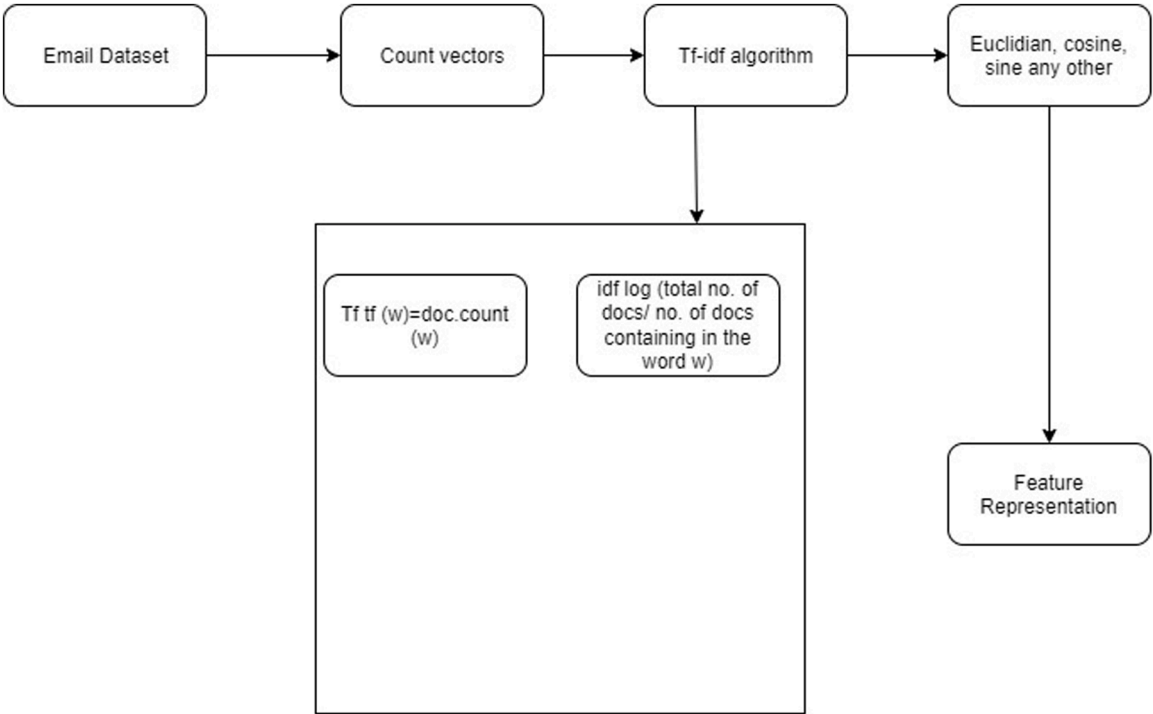


Fig. 4. TF-IDF working diagram (Grahlow et al., 2020).

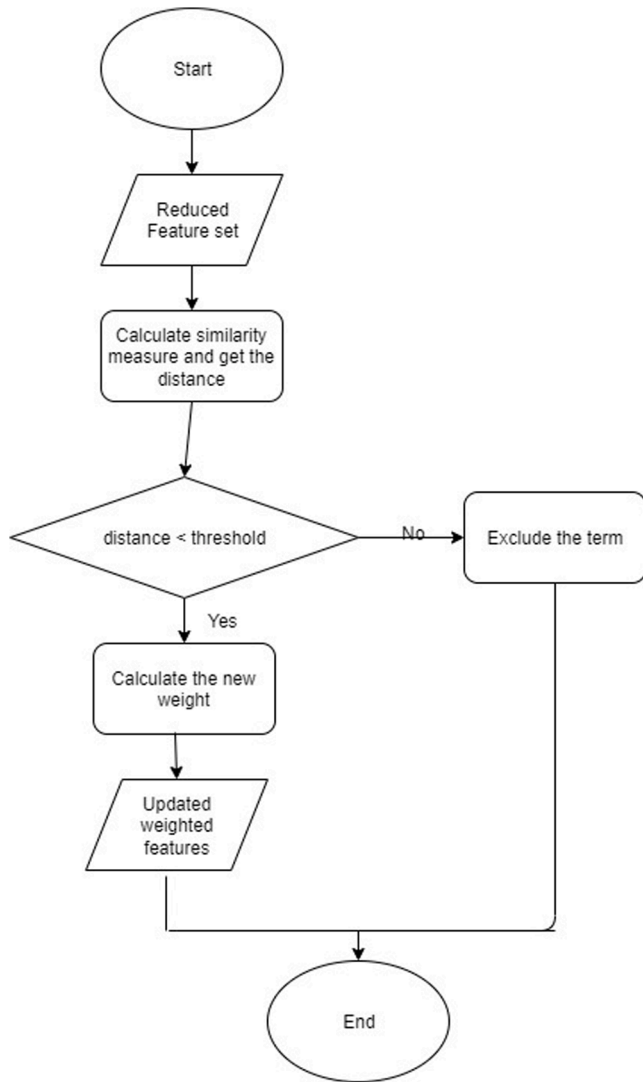


Fig. 5. Flowchart of feature weight updates process.

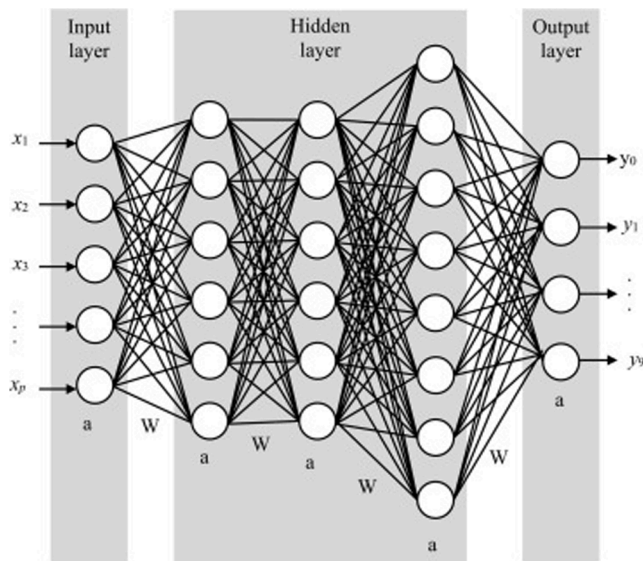


Fig. 6. Deep neural network (DNN).

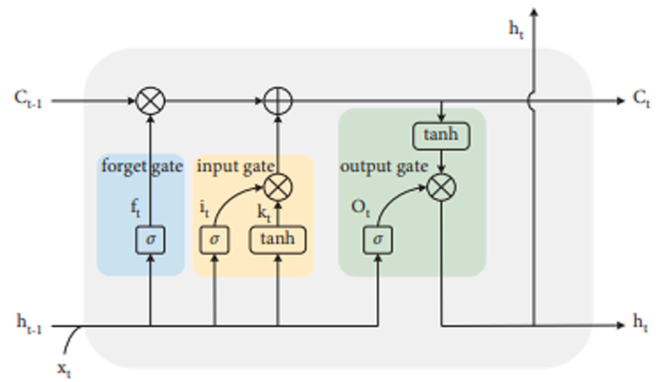


Fig. 7. Long Short-Term Memory architecture.

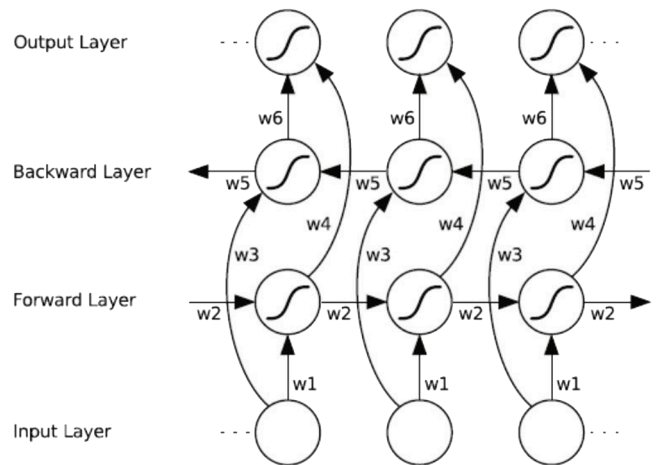


Fig. 8. Active function of BiLSTM.

with the Rabbit algorithm which is also a symmetric algorithm as AES but stream cipher in nature as additional security for data transmission. The primary goal of this post is to keep submitted data private and secure. Making sure that only authorized individuals may access data from the cloud is challenging. Then do not enable unauthorized persons to access your info. Here, we employ two distinct levels of encryption techniques. A block cipher technique is used on one layer and a stream cipher on the other. Both of the algorithms have used 256-bit and 128-bit versions of AES and Rabbit, respectively are symmetrical.

AES is considered the most powerful encryption algorithm. Interesting about AES is that it performs the computation on bytes, for example, the plaintext of 128 bits converts into 16 bytes. Depending on the length of the key size it takes rounds. Plaintext uses bytes and arranges that in the matrix in columns and rows. The first layer has used the AES-256-bit encryption algorithm. 256 bits of AES encryption uses 14 rounds and converts it to 32 bytes. In rounds, the transformation of each column is done by using mathematical functions – 1. Sub-bytes2. Shift rows, 3. Mix column, and 4. Add round key. The output completely replaces the previous column and results in new output in every round. At the last round get cipher text. While in the Decryption process, the procedure just adopts the reverse manner of the encryption algorithm. Algorithms for decryption need to be implemented separately as they are very similar to encryption processes. The rabbit algorithm here takes the cipher text from AES as input and converts it into the 128-bit text if the AES cipher text is 256-bit or 192 bits. Rabbit is a stream cipher algorithm that works only on 128-bit. It increases the high-speed performance of algorithms.

Algorithm:

Step 1: Input: Predicted Result

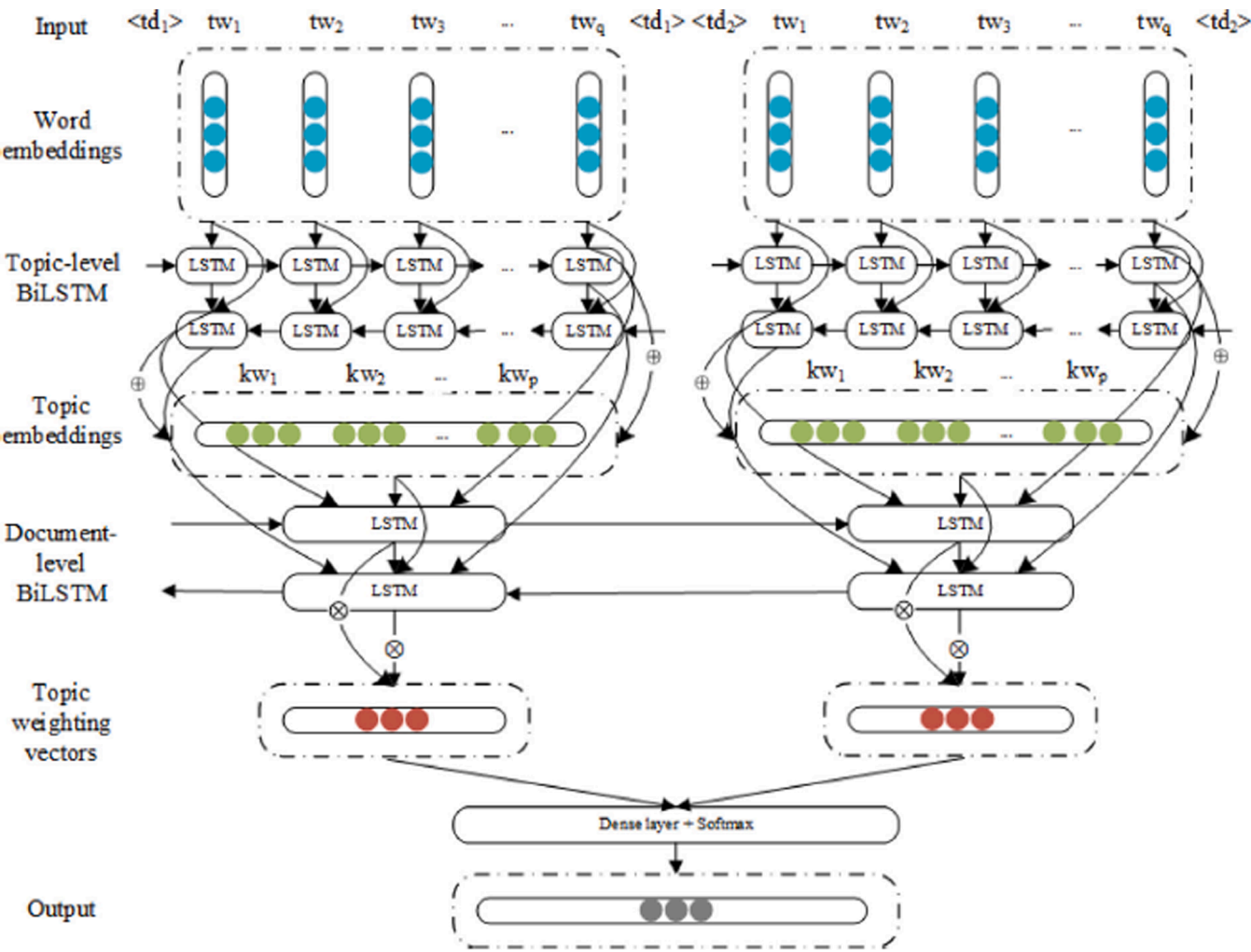


Fig. 9. Block diagram of the Proposed DNN-BiLSTM.

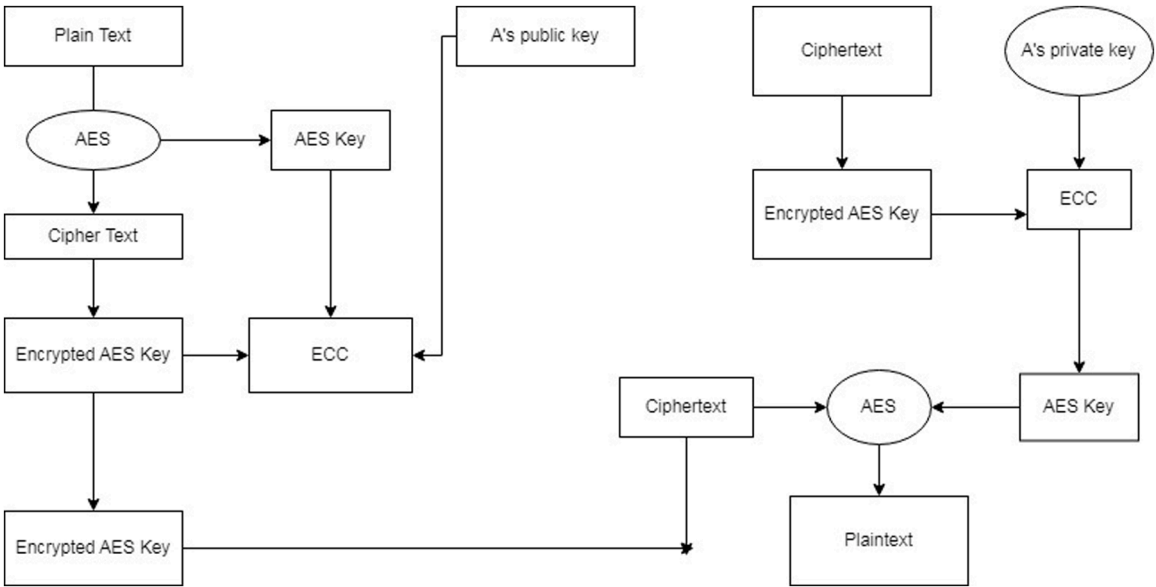


Fig. 10. Flow chart of Dual layer encryption / decryption algorithm.

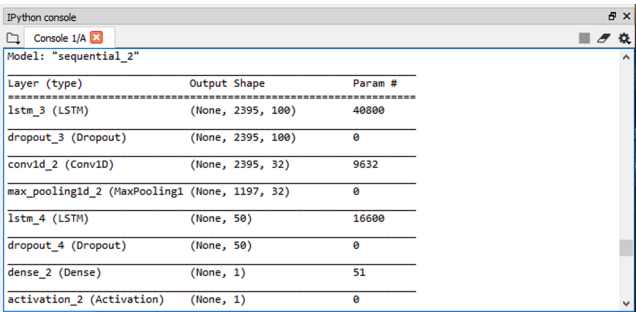


Fig. 11. Simulation Specification of DNN-BiLSTM.

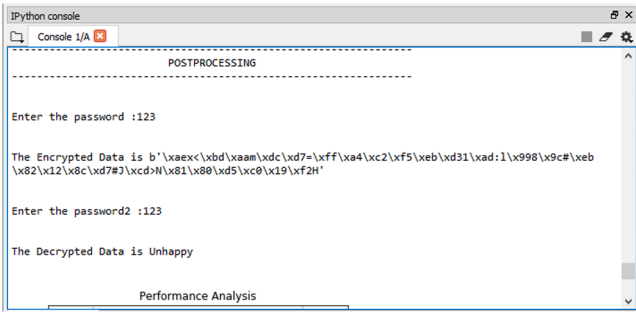


Fig. 15. Simulation results of Emotion detection in Enron dataset.

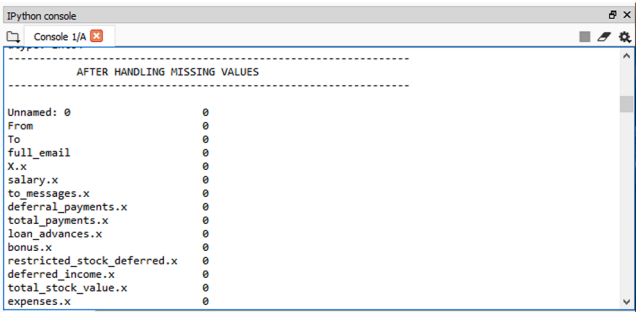


Fig. 12. Simulation results of Preprocessing.

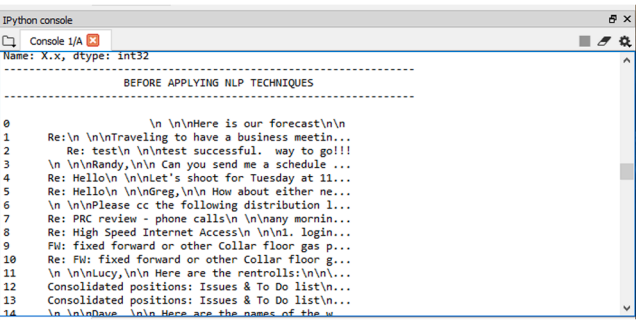


Fig. 13. Simulation results of before applying Feature Extraction of TF-IDF document.

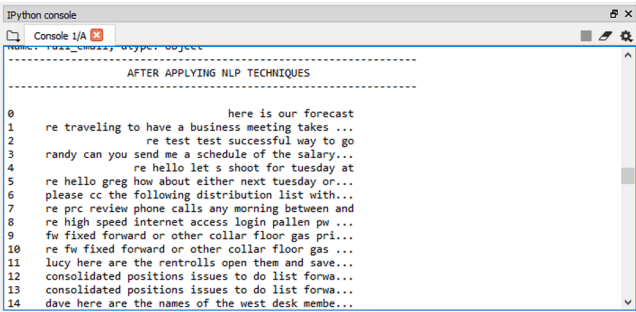


Fig. 14. Simulation results of Feature Extraction of TF-IDF document.

a. Predicted Sentiment (String) (i.e.) S1, S2,Sn

Step 2: Initialization:

a. 64 bit plain text

Table 7
AES and the rabbit algorithm's encryption and decryption times.

File Size (in KB)	Encryption (in sec)	Decryption (in sec)
1080	0.6	0.6
1919	0.9	0.9
3200	1.8	1.4
5080	2.4	2
19080	5.7	5

Table 8
Arithmetic mean height model.

	SVR	RT	RF	10-fold CV
R2	0.84	0.9	0.95	1.02
RMSE	2.17	1.78	1.50	1.23
CV RMSE	3.10	2.45	2.15	2
CV MB	-0.22	-0.24	-0.26	-0.28

Step 3: Key Generation:

- a. Round function
- b. Key schedule

Step 4: Encryption:

- a. Data is to be encrypted using Dual Encryption algorithm.
- b. String → ASCII (Ordered)
- c. Ordered → Ordered * len(string)
- d. General DES

Step 5: Decryption:

- a. Reverse of Encryption

Step 6: Output:

- a. Predicted Sentiment (String) (i.e.) S1, S2,Sn

4. Results and discussion

4.1. Experimental setup

For this research, I'm utilizing the Python PyCharm IDE and the back ends Keras and TensorFlow. On a laptop running Windows 10 (64-bit), an NVIDIA GeForce GTX 1050 graphics card, an Intel Core i7 CPU running at 2.20 GHz, and 16GB of RAM, all tests were conducted. With default settings and softmax as the output of the activation function and loss function, respectively, the loss function is a categorical cross-entropy.

4.2. Dataset

The main ingredient in all studies is data sets. The Enron dataset from the UCI repository is used in this investigation. Multiple Enron datasets total of six, which gathered. Combining all six Enron datasets resulted in the creation of the final seven Enron datasets, referred to as Combined Enron datasets. Each Enron dataset includes a set of emails made up of spam and other irregular emails that were sent and received by Enron workers. The Enron dataset was selected primarily to build a strong model capable of accurately classifying non-spam messages. Table 1 displays the complete counts for each Enron dataset.

4.3. Machine learning models

The machine learning techniques LR (logistic regression), CNN (convolutional neural network), RF (random forest), RNN (recurrent neural network), and LSTM are evaluated and compared in this paper (long-term memory). Using the Bi-LSTM approach for DNP (deep neural network) efficiency suggestions, we train machine learning models to ascertain which models are optimal for comparison and electronic forensics tools.

Results relating to the experiments are presented in this section. For experimental work, employ Python 3.8 tools on the Windows 10 operating system environment. To determine classification accuracy between spam and spam employed six Enron datasets were in this study. Two sections make up this experiment. The first section makes use of six separate Enron datasets, while the second section merges these six datasets to produce a signal-binding dataset. The number of instances in the Enron datasets is listed in Table 2.

Using preprocessing techniques including word count, steam engine, tokenizer, pruning, and silver, as shown in Table 3, we eliminate unnecessary words from both datasets. The remaining words are displayed in Table 3 following the preprocessing technique's application to the dataset. The word counts for the dataset after applying word count, simmer, tokenizer, pruning, and non-terminology preprocessing procedures are shown in Table 3.

Following this study, we employ machine learning methods to achieve the accuracy given in Table 4 such as Naive Bayes, J48, RF, Random Trees, RBF, and the suggested DNN-BiLSTM. Each dataset in Table 4 is combined with LR, CNN, RF, RNN, LSTM, and the proposed DNN-BiLSTM to reach the highest level of accuracy. Comparing the accuracy of several classifiers using various Enron datasets, Table 4 shows that the Random Forest classifier has the best accuracy.

The confusion matrix for the RF algorithm's best accuracy with the merged Enron dataset is shown in Table 5 as well.

TPR (True Positive Rate), FPR (False Positive Rate), Precision, Specificity, and F-Measure are other performance indicators that are measured. Table 6 presents the outcomes.

4.4. Emotion detection

Figs. 2, 3, 8, 10, 11, 12, 13, 14 and 15.

4.5. Delay calculation

Following a successful run and prior to the file being reloaded, the delay calculation will be performed.

$$\text{delay} = a_T - b_T \quad (33a)$$

where b_T is the time before the load and T is the time after a successful load. Experimented with various encryption and decryption delay durations (seconds) and file sizes (kB). Check out Table 7 for a comparison of encryption and decryption.

4.6. Arithmetic mean height model

Using SVR, RT, and RF, the arithmetic mean height was modelled at the parcel level (Table 8). In terms of R2, calibration RMSE, cross-validation RMSE (CV RMSE), and cross-validation MB, we compared the three models. The R2 was 0.84, the RMSE was 2.17 m, the CV RMSE was 3.10 m, and the CV MB was -0.22 m for the SVR model. Among the three models, the RT model had the lowest CV RMSE (2.45 m). However, among the three machine learning techniques, the RF model had the greatest adjusted performance, with the highest R2 value (0.95) and corrected RMSE (1.50 m).

The CV RMSE, however, is the highest among the RF models. This indicates that the model has to overfit the training set of data. RF is notoriously unsuitable, particularly when the size of the training sample is huge. The short training sample size is to blame for the substantial difference between the RF model and the CV RMSE calibration (42 plots). The calibration and cross-validation results for the RT model were comparable, but CV MB had the largest absolute value, proving that we overestimated the arithmetic mean height at the plot level. The absolute value of CV MB in the RF model was the lowest. The three models perform similarly when it comes to estimating the arithmetic mean height at the parcel level, according to the findings of the ANOVA test (all p-values for model pairings > 0.05).

5. Conclusion

A new email filtering technique is presented in this work, one that aims to enhance performance in terms of accuracy and complexity. For academics, accurately classifying spam texts is a tremendously challenging undertaking. Build a model for classifying spam documents in this post. In addition to the Enron-1, Enron-2, Enron-3, Enron-4, Enron-5, and Enron-6 datasets, we also used the merged Enron dataset. We classified spam and ham email documents in this study by analyzing the Enron dataset and applying classification techniques. Utilizing LSTM, DNN-BiLSTM, LR, CNN, RF, RNN, and suggested LSTM, create a novel model. On each of the seven datasets, the DNN-BiLSTM classifier excels over other techniques in terms of accuracy. Concerning the combined Enron dataset, the DNN-Bi-LSTM classifier offers up to 98.69 % accuracy. 88.69 % and 87.93 % accuracy were attained, respectively, by Random Forest and RNN Network classifiers. When compared to other models that were employed, these results are decent. In the future, we'll develop a hybrid model of potent computational intelligence that is more accurate than third parties at jobs like spam filtering, email categorization for phishing, and attack classification. To increase classification accuracy with fewer features and a faster calculation time, we have developed novel feature selection and optimization techniques. As a result, the dataset contains fewer nonsensical characteristics. Future research might evaluate ensemble categorization. We also investigate how the suggested semantic method behaves across various datasets. Our model is now concentrated on email author attribute manipulation for English email content; future work can be expanded to other languages, employing author confirmation processes to produce unknown authors that can be detected as unknown. Also, we create robust hybrid models that are computationally sophisticated and have excellent accuracy compared to other models for spam filtering, phishing email classification, and different forms of attacks. In order to reduce the number of pointless features in the dataset and improve classification accuracy with fewer features and faster calculation, we also create novel feature selection and optimization approaches.

Availability of information and resources

Since no datasets were produced or examined for this article, data sharing is not relevant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Governments or for-profit organisations are not permitted to finance us.

Contributions of authors

P.K. provided technical, conceptual, and architectural information. Lupus multiplex. Add to the HPP and maps. Participate in a consultation on dissertation writing.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijcce.2024.01.002](https://doi.org/10.1016/j.ijcce.2024.01.002).

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